

Instructions:

1. Create a folder with your employee ID on desktop. Save all your files in this folder only. After submitting the folder, delete it from the local machine.
2. You can refer your old assignments.
1. Marks will be only awarded to working functionalities
2. During the test switch of your mobile phone

Create **Spring Boot based REST endpoints** to **book tickets for upcoming cricket matches**

Application has following roles

- 1) Admin
- 2) Audience

Use Cases for Admin

- a) Add new Match
- b) Cricket Stadium has fixed 20 seats
- c) View Audience details by matchDate
- d) View Seat Status by matchDate

Use Cases for Audience

- a) Search upcoming matches dates
- b) Book ticket for match based on matchdate
- c) Show Booking Status by ticket Id

Below is the Class Design

```
1 package com.ds.app.test;
2
3 import java.util.List;
4
5 public class CricketMatch {
6
7     int matchNumber;
8     String dateOfMatch;
9     String cityName;
10    final int seats = 20;
11
12    List<Booking> allBookings;
13 }
14
```

```
1 package com.ds.app.test;
2
3 import java.util.List;
4
5 public class Booking {
6
7     private int bookingId;
8     private CricketMatch match;
9
10    List<Audience> allAudience;
11 }
12
```

```
*Audience.java ×
1 package com.ds.app.test;
2
3 public class Audience {
4
5     private String name;
6     private char gender;
7     private int age;
8
9     Booking booking;
10
11 }
12
```

Implement the following requirement

- a) Add new Match (5 marks)
- b) View Audience details by match Date (6 marks)
 - a. Display the Audience details in following format

```
[
  {
    seatNumber : 1,
    name : "Ramesh Kumar",
    gender : "male"
  },
  {
    seatNumber : 2,
    name : "Kavita",
    gender : "female"
  },
]
```
- c) View seat Status by match Date (5)
 - output:

```
{
  booked : 14,
  empty : 6
}
```
- d) Search upcoming matches date (5)
 - ```
{
 23-03-2026,
 27-03-2026,
 29-03-2026
}
```
- e) Book ticket for match based on match date (8)
  - a) User can book multiple tickets
  - b) Output : 4 tickets booked by "Ramesh Kumar" , seats No , 5,6,7,8 have been allocated.
- f) Show Booking Status by ticket Id (5)
  - a) Renders match date , name and seat number
- g) Add CustomException
  - a. All Fields are mandatory while booking the tickets otherwise system should render the exception "Required Field Error " + filed name ( 10)  
Note : Implement through Server Side validations
- h) Coding Standard (6)
  - a) Layered Architecture (1)
  - b) Naming Standards (1)
  - c) Endpoint standards (2)
  - d) Use of DTO (2)

Inside the resources folder, include the following:

1. A json file containing the sample request/response structures and endpoint details.
2. The application must run on port 8090.
3. Database name: cricketmatch.